

Fine TIME

Scientific Executive Summary

28 September 2026

Research question

Fine TIME asks whether task-specific fine-tuning improves pretrained time-series foundation models over their frozen zero-shot forecasts. Each task compares the same backbone before and after adaptation, using the same forecast windows and evaluation support.

Implemented experiment

The first experiment covers Chronos-2, Chronos-Bolt, and TS-ICL on the 90-task Selectime subset used by the project. Each task produces paired frozen and fine-tuned forecasts. Fine-tuning uses only model-eligible training windows with finite input and output support; it does not substitute an unfitted model when no such window exists. Seasonal Naive supplies the shared reference grid and scaling contract.

Component	Current contract
Models	Chronos-2, Chronos-Bolt, and TS-ICL
Comparison	Frozen forecast versus task-fine-tuned forecast
Task set	90-task Selectime subset
Outputs	Paired predictions, metrics, timing, manifests, and report artifacts
Run identity	Meaningful scientific path plus <code>run_n</code> ; configuration and lifecycle in <code>manifest.json</code>

Evidence boundary

The pipeline and Selena launcher are implemented, but no Fine TIME experiment has been executed yet. Therefore this document reports no accuracy, timing, or adaptation-benefit claim. The first completed run must establish whether fine-tuning improves each backbone and at what computational cost.